AP PRECALCULUS

Unit 1 • Lesson 1.10 practice • Answer key

Lesson 1.10 • Rational Functions and Holes

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Where is the hole in the graph of $r(x) = \dfrac{x^2 - 25}{x - 5}$?

(A) $(5, 0)$
(B) $(5, 10)$
(C) $(-5, 0)$
(D) $(5, 5)$

Solution

Answer: (B). $r(x) = x + 5$ for $x \neq 5$, and $5 + 5 = 10$.

Question 2

NO CALCULATOR

Where is the hole in the graph of $r(x) = \dfrac{x^2 + x - 2}{x^2 - 1}$?

(A) $\left(1, \frac{3}{2}\right)$
(B) $(-1, 0)$
(C) $(1, 0)$
(D) $(-2, 0)$

Solution

Answer: (A). $\dfrac{(x+2)(x-1)}{(x-1)(x+1)}$; the hole is at $x = 1$ with height $\frac{3}{2}$.

Question 3

NO CALCULATOR

What happens on the graph of $r(x) = \dfrac{x - 2}{(x - 2)^2}$ at $x = 2$?

(A) A hole
(B) A zero
(C) A vertical asymptote
(D) Nothing; r is defined at $x = 2$.

Solution

Answer: (C). The factor $(x - 2)$ appears more times in the denominator: $r(x) = \frac{1}{x-2}$ for $x \neq 2$.

Question 4

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 3x}{x - 3}$ is

(A) 0
(B) 1
(C) 9
(D) 3

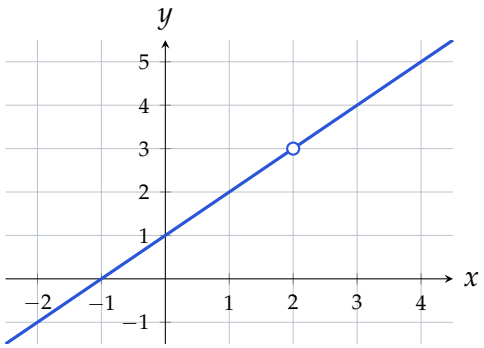
Solution

Answer: (D). $\dfrac{x(x-3)}{x-3} = x \to 3$.

Question 5 (graph)

NO CALCULATOR

The graph of a rational function r is shown: a line with a hole at $(2, 3)$. Which could be $r(x)$?



(A) $\dfrac{x^2 - 4}{x - 2}$
(B) $\dfrac{x + 1}{x - 2}$
(C) $\dfrac{x^2 - x - 2}{x - 2}$
(D) $x + 1$

Solution

Answer: (C). $\dfrac{x^2 - x - 2}{x - 2} = \dfrac{(x-2)(x+1)}{x-2} = x + 1$ for $x \neq 2$, with a hole at $(2, 3)$. Choice (A) is $x + 2$ with a hole at $(2, 4)$; (B) has an asymptote; (D) has no hole.

Question 6 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{x^3 - 4x}{x^2 + 2x}$.

(a) Factor $r(x)$ completely and simplify.
(b) Find the coordinates of every hole in the graph of r.
(c) Does the graph of r have a vertical asymptote? Find the zeros of r and describe the graph.

Solution

(a) $r(x) = \dfrac{x(x-2)(x+2)}{x(x+2)} = x - 2$ for $x \neq 0$ and $x \neq -2$.
(b) Hole at $x = 0$: $(0, -2)$. Hole at $x = -2$: $(-2, -4)$.
(c) No vertical asymptote: every factor of the denominator cancels. The only zero is $x = 2$. The graph is the line $y = x - 2$ with holes at $(0, -2)$ and $(-2, -4)$.